

AI FAQ ASSISTANT

Mini Project Report

Project Title: AI FAQ Assistant

Domain: Artificial Intelligence / Natural Language Processing

Frontend: HTML, CSS, JavaScript

Backend: Python Flask

Database: SQLite

Software: Python, VS Code, Web Browser

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1. Abstract

AI FAQ Assistant is a simple web-based application that answers frequently asked questions automatically. Users enter a question in a chat interface and the system searches a stored FAQ knowledge base to provide a suitable answer. The project reduces repeated manual support work and gives users quick access to common information. Python Flask is used for the backend, while HTML, CSS and JavaScript are used to create the user interface. SQLite can be used to store questions and answers.

2. Introduction

An FAQ system provides answers to commonly asked questions. In a traditional FAQ system, users must manually search through a list of questions. The AI FAQ Assistant makes this process easier by allowing users to type their question and receive an answer through a conversational interface.

3. Existing System

In the existing system, users usually read a static FAQ page or contact a person for support. Finding the correct information can take time, and the same questions may need to be answered repeatedly.

4. Proposed System

The proposed AI FAQ Assistant provides an interactive question-and-answer system. The user enters a question, the backend processes it, finds the closest matching FAQ, and displays the answer. New FAQ entries can be added to the database when required.

5. Objectives

- Provide quick answers to common questions.
- Reduce repeated manual support work.
- Create a simple and user-friendly interface.
- Store FAQs in an organized database.
- Demonstrate a beginner-friendly AI/NLP application.

6. Modules

1. User Interface Module – accepts user questions and displays answers.
2. Question Processing Module – cleans and processes the entered text.
3. FAQ Matching Module – compares the question with stored FAQ questions.
4. Database Module – stores FAQ questions and answers.
5. Response Module – returns the most suitable answer to the user.

7. Technologies Used

Frontend: HTML, CSS, JavaScript.

Backend: Python with Flask.

Database: SQLite.

AI/NLP: keyword matching or text similarity can be used for FAQ matching.

Editor: VS Code or any Python-supported IDE.

8. System Flow

User → Enter Question → Flask Backend → Question Processing → FAQ Matching → Database/Knowledge Base → Answer → User

9. Advantages

- Fast response to common questions.
- Easy to use.
- Reduces repetitive work.
- Low-cost implementation.
- Easy to update FAQ data.
- Can be extended with advanced NLP models.

10. Limitations

- Basic matching may fail for unfamiliar questions.
- The quality of answers depends on the FAQ database.
- It may not understand complex conversations without advanced NLP.

11. Future Enhancement

The system can be improved by adding machine-learning or large-language-model based responses, voice input, multilingual support, login functionality, an admin panel, analytics, and a larger knowledge base.

12. Conclusion

AI FAQ Assistant is a useful beginner-level AI project that demonstrates how a web application can automatically answer frequently asked questions. It combines a simple frontend, Python Flask backend and database, and can later be enhanced with more advanced natural-language processing.

13. Sample Project Structure

```
ai_faq_assistant/  
├── app.py  
├── database.db  
├── templates/  
│   ├── index.html  
│   └── static/  
│       ├── style.css  
│       └── script.js  
└── README.md
```

14. Sample FAQ Data

Question	Answer
What is this system?	It is an AI FAQ Assistant that answers common questions.
How does it work?	It matches the user's question with available FAQ information.
Which technology is used?	Python Flask, HTML, CSS, JavaScript and SQLite are used.
Can FAQs be updated?	Yes, the FAQ database can be updated with new questions and answers.